

OLIST BRAZILIAN E-COMMERCE DATA ANALYSIS

Overview

This project analyzes the **Olist Brazilian E-Commerce Dataset from Kaggle**, a real-world marketplace dataset containing customer purchases, product information, payments, and order details from a Brazilian online retail platform.

The analysis combines **5 relational tables**:

- **orders**: Order tracking, timestamp data, and transaction statuses.
- **order_items**: Item counts, pricing details, and product IDs per transaction.
- **order_payments**: Total transaction payment values and payment methods.
- **products**: Mapping of product IDs to product category names.
- **customers**: Mapping of transaction-level `customer_id` keys to individual account `customer_unique_id` keys.

Date Range: September 2016 – October 2018

Data Quality Notes

- `customer_unique_id` was used instead of transaction-level `customer_id` to accurately track individual customers and repeat purchases.
- Records with missing product category labels were excluded from category-level rankings.
- Revenue analysis was based on product sales values from the available order data.

Business Questions

1. Who are the top 10 customers by total amount spent?
 2. What is the monthly revenue trend across the dataset?
 3. What is the month-over-month (MoM) change in revenue?
 4. Which product categories generate the most revenue?
 5. What are the top 3 products within each category ranked by revenue?
 6. How can customers be segmented into Low, Medium, and High Spend tiers?
 7. How many customers are repeat buyers versus one-time buyers?
 8. What percentage of total revenue comes from the top product category?
-

Insight Summary

Severe Retention Gap

The customer retention analysis found that **96.9% of customers (93,098 buyers) made only one purchase**, while only **3.1% (2,997 buyers) made repeat purchases (Query Result 7)**. This indicates a significant opportunity to improve customer retention and increase customer lifetime value.

Strong Seasonal Demand

Monthly revenue peaked at **\$1,194,882.80 in November 2017**, representing a **\$415,204.92 month-over-month increase (Query Result 3)**. This indicates that seasonal shopping periods and promotional events have a significant impact on sales performance.

High-Value Customers Drive Revenue

Customer segmentation found that approximately **10% of customers belong to the High Spend Tier (≥\$300)** while generating approximately **42% of total revenue (Query Result 6)**. This shows that a relatively small customer segment contributes a disproportionately large share of marketplace sales.

Diversified Revenue Structure

Cama_mesa_banho or **bed_table_bath** was the highest-revenue category, generating **\$1,712,553.67**, or **10.7% of total revenue (Query Result 4)**. Since no single category dominates the marketplace, revenue is relatively diversified across product categories.

Key Findings

The Olist marketplace generated approximately **\$16 million in total revenue** during the analysis period. Revenue grew significantly over time, reaching its highest monthly value of **\$1.19 million in November 2017**.

Customer analysis revealed **96,095 unique customers** with an average spend of **\$160.99**. Approximately **70%** of customers were classified as Low Spend, **20%** as Medium Spend, and **10%** as High Spend.

The largest business challenge is customer retention, with **96.9% of buyers making only one purchase**. At the same time, high-value customers contribute a substantial share of revenue, creating an opportunity for targeted retention and loyalty strategies.

Product revenue was led by **Cama_mesa_banho** or **bed_table_bath (\$1.71M)**, followed by **beleza_saude** or **beauty_health (\$1.66M)**** and ****informatica_acessorios** or **IT accessories (\$1.59M)**. The top category represented only **10.7% of total revenue**, indicating a relatively diversified product mix.

Strategic Recommendations

1. **Improve Customer Retention:** Deploy targeted 30-day email and cross-sell campaigns to address the **93,098 one-time buyers (Query Result 7)** and encourage first-time customers to make repeat purchases.
 2. **Establish a High-Tier VIP Program:** Provide dedicated support, bundling incentives, and loyalty rewards for High-Tier customers (**$\geq \$300$**), who represent approximately **10% of customers but generate ~42% of total revenue (Query Result 6)**.
 3. **Plan for Seasonal Demand:** Increase inventory and marketing capacity ahead of peak periods to capture the **+\$415,204.92 November revenue increase (Query Result 3)**, while adjusting December spending and inventory based on the observed post-peak decline.
-

Tools & Skills

Tools

- MySQL
- DataGrip

SQL Skills

- Common Table Expressions (CTEs)
- Window Functions
- Aggregate Functions
- Conditional Logic
- Relational Joins
- Data Cleaning
- Revenue Analysis
- Customer Segmentation
- Product Ranking